

Symphonym: Universal Phonetic Embeddings for Cross-Script Name Matching

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Abstract

Linking names across historical sources, languages, and writing systems remains a fundamental challenge in digital humanities and geographic information retrieval. Existing approaches require language-specific phonetic algorithms or fail to capture phonetic relationships across different scripts. This paper presents Symphonium, a neural embedding system that maps names from any script into a unified 128-dimensional phonetic space, enabling direct similarity comparison without runtime phonetic conversion. Symphonium uses a Teacher-Student architecture where a Teacher network trained on articulatory phonetic features produces target embeddings, while a Student network learns to approximate these embeddings directly from characters. The Teacher combines Epitran (extended with 100 new language-script mappings), Phonikud for Hebrew, and CharsiuG2P for Chinese, Japanese, and Korean. Training used 32.7 million triplet samples of toponyms spanning 20 writing systems from GeoNames, Wikidata, and Getty Thesaurus of Geographic Names. On the MEHDIE Hebrew-Arabic historical toponym benchmark, Symphonium achieves Recall@10 of 97.6% and MRR of 90.3%, outperforming Levenshtein and Jaro-Winkler baselines (Recall@1: 86.7% vs 81.5% and 78.5%). Evaluation on 12,947 real cross-script training pairs shows 82.6% achieve ≥ 0.75 cosine similarity, with best performance on Arabic-Cyrillic (94–100%) and Cyrillic-Latin (94.3%) combinations. The fixed-length embeddings enable efficient retrieval in digital humanities workflows, with a case study on medieval personal names demonstrating effective transfer from modern place names to historical orthographic variation.

Keywords: phonetic embeddings; cross-script matching; teacher-student distillation; neural networks; digital humanities; name matching

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1 Introduction

Historical gazetteers aggregate place references from sources spanning millennia, dozens of languages, and multiple writing systems. The World Historical Gazetteer¹ (WHG) alone indexes over 67 million toponym records from antiquity to the present, drawn from sources as diverse as classical Latin itineraries, medieval Arabic geographies, and modern national mapping agencies. Each toponym record is tagged with language and script metadata, though the same surface form (e.g., “London”) may appear multiple times when attested in different languages.

The core difficulty is that place name similarity is fundamentally *phonetic*, not orthographic. English speakers recognise “København” and “Copenhagen” as the same city not because the spellings match, but because the sounds are similar enough to suggest common origin. Yet computational approaches to toponym matching have largely relied on string-based metrics (edit distance, Jaro-Winkler) or phonetic algorithms designed for single languages (Soundex, Metaphone for English; Cologne phonetic for German). These approaches fail catastrophically when names cross script boundaries: no amount of edit distance tuning will link “東京” to “Tokyo”.

Recent work has applied neural embeddings to geographic entity matching (Qiu et al., 2024; Rama, 2016; Sagi et al., 2025), but existing systems either operate within single scripts, require language identification at inference time, or depend on curated transliteration resources that don’t scale to the long tail of historical languages. Until now, there has been no system that can place “Νέο Μεξικό” (Greek), “নিউ মেক্সিকো” (Bengali), “نيومكسيكو” (Arabic), and “Нью-Мексико” (Cyrillic) near each other in embedding space using only the raw character sequences as input.

This paper presents such a system, which I call *Symphonym*. The key contributions are:

1. A **script-aware but script-agnostic embedding architecture** that maps toponyms from 20+ writing systems into a unified 128-dimensional space where proximity reflects phonetic similarity.
2. A **Teacher-Student training framework** that transfers phonetic knowledge from articulatory features derived from IPA transcriptions (via Epi-tran, Phonikud, and CharsiuG2P) to a character-level model requiring no phonetic resources at inference time.

¹<https://whgazetteer.org>

3. A **noise-aware training regime** with strategic language dropout that forces the model to learn script-intrinsic phonetic patterns rather than relying on language-specific shortcuts.
4. **Phonetic similarity filtering** for training pairs using PanPhon feature distance to prevent the model from learning false equivalences between unrelated exonyms (e.g., “Germany” $\neq$ “Deutschland”). This filtering ensures training samples represent genuine phonetic correspondences—variants that sound alike—rather than semantic or cultural name substitutions. The filtering improves model precision without biasing toward particular scripts or languages, as it operates on universal articulatory feature space.
5. A **three-phase curriculum** progressing from phonetic feature learning through knowledge distillation to hard negative discrimination.
6. **Empirical validation** of cross-domain transfer: a case study on medieval personal name matching (Section 8) demonstrates that embeddings trained on modern toponyms transfer effectively to historical personal names with pre-standardisation orthography.

The resulting system will enable fuzzy phonetic search across the full WHG corpus without language identification, script-specific preprocessing, or runtime phonetic conversion. Beyond cross-script matching, the phonetic embedding approach naturally handles *historical orthographic variation*—the inconsistent spellings characteristic of pre-standardisation documents—by grouping sound-alike variants near their modern canonical forms. This capability extends beyond toponyms to personal names, and in any textual domain involving phonetically consistent but orthographically variable name forms.

Scope and Integration. This work addresses the *name-matching* component of toponym resolution specifically. The model has no access to geographic coordinates or spatial context—it operates purely on phonetic similarity between strings. Both Symphonium and WHG’s new reconciliation pipeline (termed WHG PLACE: Place Linkage, Alignment, and Concordance Engine) build on architecture developed by Grossner and Mostern (2021). Within PLACE, phonetic similarity serves as one input to a larger process: candidate toponyms are first retrieved via the phonetic-aware toponym index, then places containing matching toponyms are selected from the places index and filtered by geographic proximity and other contextual criteria. The contribution here is the phonetic similarity component alone.

In practical terms, Symphonium will enhance WHG’s Reconciliation Service API² (v2.0), facilitating the work of digital humanities and GLAM (galleries, libraries, archives, museums) professionals who seek to link place references in

²<https://docs.whgazetteer.org/content/technical/apis.html#reconciliation-service-api>

historical documents, archival catalogues, and research datasets to WHG’s consolidated index. Once linked, these references become entry points to a web of related materials—other sources attesting the same places, variant historical names, and cross-references across collections—even when the original materials span multiple scripts, languages, and historical periods.

2 Related Work

Classical and String-based Approaches. Early phonetic algorithms—Soundex (Russell, 1918), Metaphone (Philips, 1990), and PHONIX (Gadd, 1990)—rely on hand-crafted rules for specific languages. Trevor Gadd’s PHONIX encoded 160 transformation rules for Latin-script Southern African languages, whereas Symphonym learns correspondences from 32.7 million samples across 20 scripts. Edit distance metrics (Levenshtein, Jaro-Winkler) measure orthographic similarity and fail across scripts. While Recchia and Louwerse (2013) and Santos et al. (2018) show that combining multiple string metrics improves within-script matching, these cannot generalise cross-script.

Neural Entity Matching. Recent neural approaches operate within constrained scopes. Qiu et al. (2024) achieve strong Chinese address matching using BERT with Chinese-specific features (Pinyin, radicals), but these lack analogues in other scripts. MEHDIE (Sagi et al., 2025) addresses Hebrew-Arabic historical toponym matching, demonstrating that direct script-to-script approaches can outperform romanisation for related languages, though requiring manual rules per language pair. Rama (2016) applies Siamese networks to cognate identification from character-level representations. Purely data-driven approaches must discover phonetic relationships through co-occurrence alone; Symphonym’s articulatory feature grounding explicitly encodes these relationships for more efficient learning and better low-resource generalisation.

Phonetic Grounding. The PWESUITE evaluation (Zouhar et al., 2024) demonstrates that models incorporating articulatory features show improved cross-language transfer compared to IPA-only or character-level inputs, and that phonetically grounded models trained on one language can generalise to others. These findings inform my Teacher network design. However, PWESUITE addresses monolingual embeddings within single scripts; my contribution extends validated phonetic grounding to multi-script settings via Teacher-Student distillation. Siamese networks (Bromley et al., 1993; Chopra et al., 2005) provide the architectural foundation for similarity learning, with applications from face verification (Schroff et al., 2015) to text matching (Mueller and Thyagarajan, 2016).

Cross-lingual Representations. The cross-lingual embedding literature (Conneau et al., 2018; Artetxe et al., 2018) targets word meaning rather than phonetic

form. TRANSLIT (Benites et al., 2020) provides large-scale name variants for transliteration training, capturing conventionalised mappings. Symphonism’s native script processing—CJK via CharsiuG2P (Zhu et al., 2022), Hebrew via Phonikud (Kolani et al., 2025), Cyrillic/Arabic/Greek via extended Epitran—sidesteps romanisation variance by operating in IPA-derived phonetic space. PanPhon (Mortensen et al., 2016) maps IPA to articulatory features.

3 Architecture

My system employs a Teacher-Student architecture where a Teacher network, trained on articulatory phonetic features, produces target embeddings that a Student network learns to approximate from character sequences alone. At inference time, only the Student is required, enabling deployment without phonetic resources. This design addresses the limitations of prior approaches discussed in Section 2, particularly the reliance on language-specific phonetic rules and single-script operation.

3.1 Design Principles

Three principles guide my architecture:

Script Awareness Without Script Dependence. The model must handle 20 writing systems but produce embeddings in a unified space where script boundaries are transparent. I achieve this through deterministic script detection (via Unicode block analysis) combined with learned script embeddings that allow the model to interpret characters appropriately for each script.

Phonetic Grounding. Embedding similarity must reflect phonetic similarity, not orthographic or semantic similarity. I ground the embedding space through the Teacher network, which operates on articulatory features derived from IPA transcriptions using PanPhon.

Inference Efficiency. The deployed model must encode toponyms without runtime phonetic conversion, language identification, or external resources. All phonetic knowledge is distilled into the Student’s parameters during training.

3.2 Architecture Overview

Figure 1 illustrates the Teacher-Student architecture and the three-phase training curriculum.

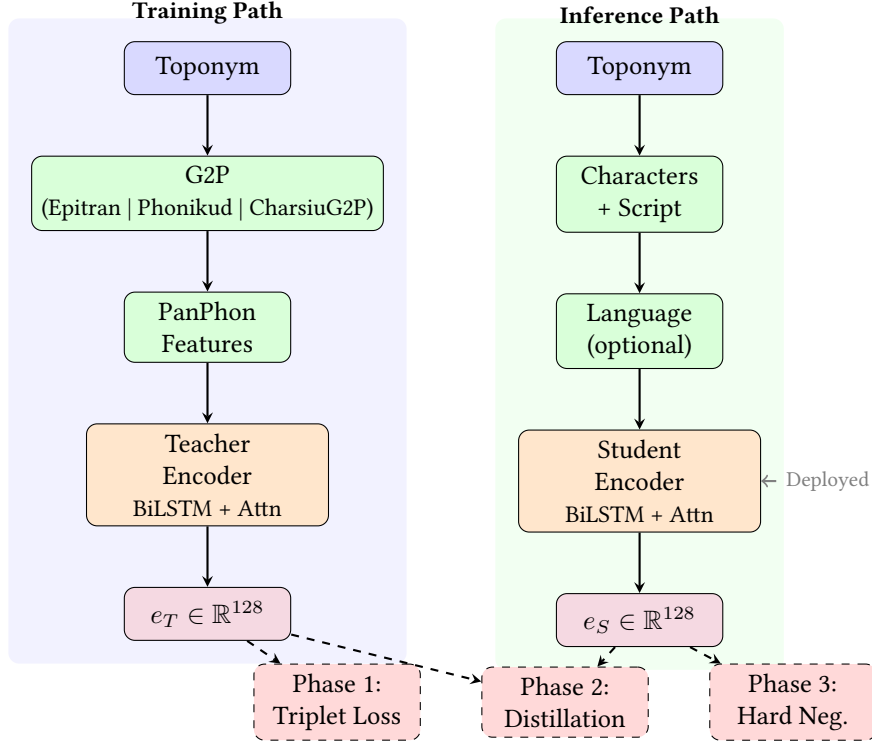


Figure 1: Teacher-Student architecture. **Left:** The Teacher pathway converts toponyms to IPA, extracts PanPhon articulatory features, and encodes to 128-d embeddings. Used only during training. **Right:** The Student pathway processes raw characters with script/language metadata. Deployed at inference time without phonetic resources. **Bottom:** Three training phases progressively transfer phonetic knowledge from Teacher to Student.

3.3 Script Detection and Character Vocabulary

I detect script from Unicode code points, mapping each character to one of 20 script categories. The vocabulary contains **113,280 tokens** observed across 66.9 million toponyms: approximately 5,300 alphabetic characters (Latin, Cyrillic, Arabic, Greek, and other scripts), 93,549 observed Han characters from the CJK Unified Ideographs block, 11,624 Hangul syllables, and 2,817 characters from other scripts. Japanese Kanji are treated as CJK characters, while Hiragana and Katakana are classified as separate script categories. Korean Hangul syllables are decomposed into 51 Jamo components for the Student network’s character embeddings, providing explicit phonetic structure. For IPA transcription, Chinese topolects (Mandarin, Gan, Wu, Yue) and Korean are processed via CharsiuG2P, and Hebrew via Phonikud. The current version of Symphony (v6) incorrectly routes all Japanese (`lang=ja`) toponyms to CharsiuG2P (which only processes Kanji), despite native Epitran support via `jpn-Hira` and `jpn-Kana` codes (see §9). Each

token is tagged with its source script, and the vocabulary supports 1,944 distinct language codes observed in the corpus.

3.4 Teacher Network: Phonetic Feature Encoder

The Teacher network encodes toponyms via IPA transcriptions and articulatory features (Figure 1, left). The architecture comprises: feature projection to hidden dimension, bidirectional LSTM, multi-head self-attention, learned attention pooling, and output projection to 128 dimensions with L2 normalisation. The Teacher is trained on triplets of co-located toponyms, learning to place phonetically similar names (same place, different languages) closer than dissimilar names (different places).

3.5 Generalisation Through Articulatory Features

Grounding the embedding space in PanPhon articulatory features enables generalisation to languages with limited training representation. PanPhon represents each IPA segment as a 24-dimensional binary vector encoding universal articulatory properties (place, manner, voicing), independent of language identity. The phoneme /b/ receives identical features whether appearing in English “Berlin”, Russian “Берлин”, or Arabic “برلين”—all encode a voiced bilabial stop. When the Teacher learns that toponyms with similar PanPhon feature sequences should have similar embeddings, it learns relationships between articulatory configurations, not specific languages. This knowledge transfers automatically to any script representing the same sounds, even those absent from training. For scripts with limited Student coverage, the Teacher network can be used directly at inference if IPA transcription is available—valuable for historical or rare scripts.

3.6 Student Network: Universal Character Encoder

The Student network processes raw character sequences with script and language metadata (Figure 1, right). Each character maps to a learned 96-dimensional embedding via script-partitioned vocabulary lookup. Script (deterministically detected) and language (ISO 639 code when available, otherwise <UNK>) contribute 16-dimensional embeddings concatenated to character embeddings. During training, known languages are randomly replaced with <UNK> (50% probability), forcing the model to learn script-intrinsic patterns. The architecture mirrors the Teacher: BiLSTM, self-attention, attention pooling, and output projection to 128 dimensions with L2 normalisation.

3.7 Noise Augmentation

To prepare the model for handling OCR errors, historical spelling variations, and transcription inconsistencies, I apply character-level noise during Student training (Phases 2 and 3): insertions, deletions (probability 0.1 each), substitutions

and transpositions (probability 0.05 each), using characters from the same script. Noise is applied with 30% probability per training example, with the target being the Teacher’s embedding of the *clean* input, training the Student to map corrupted inputs to the same phonetic region as their clean counterparts.

4 Training Data

4.1 Data Selection and Curation Criteria

I extract training data from WHG’s `places` index (geographic entities) and `toponyms` index (name variants with script/language metadata), restricting to three major namespace authorities—GeoNames (`gn`), Wikidata (`wd`), and the Getty Thesaurus of Geographic Names (`tgn`)—while excluding less rigorously-curated sources. Wikidata inclusion is critical for non-Latin script coverage, particularly South Asian (Devanagari, Tamil, Telugu, Kannada, Malayalam, Bengali) and Middle Eastern (Arabic, Hebrew) scripts where GeoNames and TGN lack sufficient multilingual correspondences. Training pairs are generated from co-located name attestations: when an authority lists multiple variants (e.g., “London”, “Londres”, “Лондон”) within the same place record, I construct triplets as described in Section 3. To address script and language imbalances, I define four selection criteria:

1. Script-Language Stratified Sampling. To prevent dominance by Latin script and European languages, I implement quota-based sampling by script+language pair (e.g., `LATIN:en`, `CYRILLIC:ru`), ensuring representation of under-resourced languages within well-resourced scripts. Each bin is capped at 50,000 samples; bins below 1,000 are dropped, while small bins (1,000–50,000) may be oversampled up to $5\times$. This prevents high-resource bins from dominating gradient signal while ensuring coverage across all script-language combinations.

2. Global Vocabulary Construction. To prevent out-of-vocabulary errors at inference, I decouple vocabulary construction from pair generation. I scan the entire corpus (66.9M unique toponyms) to build the character vocabulary, observing 113,277 distinct characters across 20 scripts. Korean Hangul syllables are retained natively (11,624 observed), CJK ideographs selectively (93,549 of 20,900 in block `U+4E00–U+9FFF`), and other scripts in native form. Including special tokens, the vocabulary contains **113,280 tokens**. Training pairs are then generated from co-located toponyms, restricted to stratified quotas.

3. Cross-Script Pairs via Clustering. Cross-script pairs (e.g., 北京 vs. Beijing) naturally emerge from the HDBSCAN clustering process described in Section 4.4. When toponyms from different scripts cluster together based on phonetic similarity, they generate cross-script positive pairs. This data-driven approach captures

genuine phonetic relationships rather than relying on heuristic weighting, ensuring the model learns from pairs that are phonetically meaningful regardless of their scripts.

4. Place-Local Deduplication. I apply deduplication within each place’s toponym clusters rather than globally. The same toponym pair may appear across multiple authorities (e.g., GeoNames, Wikidata, and TGN all list London/Londres). Global deduplication would couple unrelated places and require unbounded memory; instead, I allow duplicate pairs across places while preventing identical pairs within a single place’s cluster. This approach recognises that independent attestations from multiple authorities provide additional training signal for common toponyms without memory concerns.

4.2 IPA Transcription and Feature Extraction

The Teacher network requires IPA transcriptions generated via three complementary backends: Epitran (Mortensen et al., 2018) (extended to 100 languages), Phonikud for Hebrew, and CharsiuG2P for Chinese topolects (Mandarin, Gan, Wu, Yue) and Korean. Japanese Kana scripts are handled incorrectly in v6 (see §9).

PanPhon converts IPA segments into 24-dimensional articulatory feature vectors (voicing, place, manner, etc.). To obtain fixed-length representations from variable-length IPA sequences, I apply 8-bin position pooling: each sequence is divided into 8 positional bins (capturing word shape: start, middle, end), and the 24 articulatory features per segment are averaged within each bin, yielding $8 \times 24 = 192$ dimensions. This Symphonism-specific transformation (which I will refer to as PanPhon192) preserves positional information while mapping variable-length sequences to fixed representations suitable for clustering and dense vector storage. This fixed bin count creates length-dependent representation quality, with short strings producing sharper, more distinctive phonetic profiles while long strings average features across many segments per bin (discussed further in Section 7.3). Names without supported G2P are excluded from Teacher training but processed by the Student, which generalises from related scripts.

4.3 Attestation-Based Negative Validation

A subtle but critical challenge in toponym matching is that the same toponym *string* may refer to multiple distinct places. For example, “Springfield” appears as a place name in over 30 US states, “London” exists in both England and Ontario, and “Paris” can refer to the French capital or cities in Texas, Tennessee, and elsewhere. Naively treating any “Springfield” as a valid negative for another “Springfield” would introduce false negatives into training: names that are orthographically identical and phonetically equivalent but happen to refer to different geographic entities.

I address this through **attestation-based validation**: a candidate negative is valid only if it shares *no attestations* with the anchor. This ensures that even if “Springfield” appears in both the anchor and negative candidate pools, the two instances will not be paired as negatives if they share any common place reference. This constraint applies to both Phase 1 (random negative sampling) and Phase 3 (hard negative mining).

4.4 Phonetic Clustering for Pair Generation

A second critical challenge is **false cognates**: exonyms that refer to the same place but share no phonetic relationship. For example, “Germany” and “Deutschland”, or “日本” (Nihon) and “Japan”, denote the same entity but should not be trained as phonetically similar. Naively including such pairs would corrupt the embedding space.

Rather than applying arbitrary similarity thresholds, I use density-based clustering (HDBSCAN) on the PanPhon192 embeddings to identify phonetically coherent groups within each place’s toponyms. For each place record with multiple name variants, I:

1. Extract PanPhon192 embeddings for all toponyms with valid IPA transcriptions
2. Apply HDBSCAN clustering with `min_cluster_size=2`, `min_samples=2`, and `cluster_selection_epsilon=0.2` (the epsilon parameter merges clusters within cosine distance 0.2, i.e., similarity ≥ 0.8)
3. Generate positive pairs only from toponyms within the same cluster

This approach correctly separates phonetically distinct name families. For example, a place record for Cologne might contain:

- **Cluster 1 (Germanic)**: Köln (de), Keulen (nl)
- **Cluster 2 (Romance/English)**: Cologne (en/fr), Colonia (es/it), Colonia (la)

Pairs are generated within clusters (Köln–Keulen, Cologne–Colonia) but not across them (Köln–Cologne), preventing the model from learning that phonetically unrelated exonyms should cluster together.

Multi-script Clustering Examples. A **London** place record (wd:Q1001456) with 21 multilingual toponyms yields a single dominant cluster of 17 members spanning Arabic, CJK, Cyrillic, and Latin scripts (intra-cluster similarity 0.91): London (de, vi, pl, hu, cs, es, it, tr, nl, sv, en), لندن (fa, ar, ur), Лондон (ru, uk), and 伦敦 (zh) cluster together, while French/Portuguese London (fr, pt), Bengali লন্ডন, and Serbian Ландон are correctly isolated as phonetically distinct variants.

Similar multi-cluster patterns emerge for Moscow (2 major clusters), Beijing (3 clusters for “Beijing”, “Peking”, “Pechino”), and Paris (3 clusters across 7 scripts).

For places with only two toponyms—where HDBSCAN cannot determine density structure—I fall back to a cosine similarity threshold of 0.5 on the PanPhon192 embeddings.

4.5 Dataset Statistics

The full WHG places index contains 47.1 million place records, from which I extract 112.0 million toponym records spanning 1,944 languages and 20 script categories. During extraction, 1.77 million toponyms (1.6%) are filtered as “pre-romanised” forms—cases where the language tag implies a non-Latin script but the name is written in Latin characters (e.g., Chinese names romanised to Pinyin without explicit tagging). These are excluded to prevent the model from learning spurious orthographic similarities. After deduplication by toponym identifier, **66.9 million unique toponyms** remain.

The training subset from the three high-quality namespaces yields 57.6 million toponyms with the script distribution shown in Table 1.

IPA Coverage by Language. The 29.0M toponyms with valid IPA transcriptions span 34 script:language combinations. Top contributors include: Latin:en (8.0M), Latin:fr (2.3M), Latin:nl (2.3M), Latin:de (2.1M), Latin:sv (1.7M), Latin:es (1.5M), CJK:zh (1.3M), plus substantial coverage in non-Latin scripts: Arabic:fa (577K), Arabic:ar (412K), Cyrillic:uk (436K), Cyrillic:ru (0.8M), Hangul:ko (229K), Thai:th (210K).

Training Pair Generation. From 8.2 million places with at least two toponyms, HDBSCAN clustering generates **65.1 million positive pairs** across **595 script:language bins**. The top bins are: LATIN:en|LATIN:nl (1.49M), LATIN:de|LATIN:en (1.31M), LATIN:en|LATIN:fr (1.05M), LATIN:en|LATIN:es (0.98M).

To prevent dominance by high-resource combinations, bins are balanced: 264 (44%) exceed the 50k cap and are downsampled; 329 (55%) are oversampled up to $5\times$ their original size with stochastic negative variation to prevent memorisation. This yields **27.6 million balanced pairs** for triplet generation, representing the largest known phonetically-supervised corpus for cross-script toponym matching (cf. TRANSLIT’s 3M pairs (Benites et al., 2020); MEHDIE’s evaluation-scale hundreds).

From these pairs, I generate triplets for contrastive training:

- **Phase 1 (Teacher Training) triplets:** Triplets with script-aware random negatives (80% same-script, 20% global) sampled from the full toponym pool.

Script	Count	%	IPA Coverage	Top Languages (IPA)
LATIN	55,617,677	83.1%	49.8%	en, fr, nl, de, sv
CYRILLIC	3,614,762	5.4%	47.1%	ru, uk, bg, sr
CJK	2,973,525	4.4%	50.1%	zh (CharsiuG2P)
ARABIC	2,098,089	3.1%	52.5%	fa, ar, ur
HANGUL	393,996	0.6%	58.0%	ko (CharsiuG2P)
OTHER	342,642	0.5%	0.0%	—
KATAKANA	340,555	0.5%	0.0%	ja (Epitrans)*
THAI	251,458	0.4%	83.6%	th
GREEK	217,997	0.3%	77.4%	el (Epitrans ext.)
DEVANAGARI	166,957	0.2%	57.2%	hi, mr, ne
ARMENIAN	153,467	0.2%	93.7%	hy (Epitrans ext.)
HIRAGANA	151,980	0.2%	0.0%	ja (Epitrans)*
HEBREW	151,960	0.2%	83.8%	he (Phonikud)
BENGALI	106,896	0.2%	72.9%	bn
GEORGIAN	105,902	0.2%	81.2%	ka
MALAYALAM	68,176	0.1%	78.5%	ml
TAMIL	52,486	0.1%	90.9%	ta
TELUGU	51,440	0.1%	92.6%	te
KANNADA	43,155	0.1%	48.6%	kn (Epitrans ext.)
GUJARATI	21,428	0.03%	94.9%	gu (Epitrans ext.)

Table 1: Script distribution across all 66.9M unique toponyms. IPA coverage percentages reflect successful transcription within the 57.6M training namespace toponyms (gn/wd/tgn). v6 achieves 53.4% overall IPA coverage (30.8M toponyms) via Epitrans extensions (102 language-script pairs), Phonikud (Hebrew), and CharsiuG2P (CJK, Korean). * Both Hiragana and Katakana have 0% IPA coverage due to a routing bug (see §9 for details and observed robustness).

- **Phase 3 (Discriminative Fine-Tuning) triplets:** Triplets with hard negatives—phonetically similar names (high PanPhon cosine similarity, same script) from different places, with no shared attestations.

4.6 Training Pipeline Infrastructure

The complete data pipeline processes 67M toponyms across 20 scripts using DuckDB for analytical queries and incremental checkpointing at each major stage (positive pairs, training samples, triplets). Long-running jobs on shared HPC clusters checkpoint progress to persistent storage, enabling resumption after interruptions.

Training was conducted on University of Pittsburgh’s CRC cluster using NVIDIA L40S GPUs: Phase 1 (50 epochs, 33.5h, final val_loss 0.0056), Phase 2 (50 epochs, 1.5h, final val_loss 0.0575), Phase 3 (30 epochs, 7.5h, final val_loss 0.0216). Embedding inference for 67M toponyms required 2.5h. Final embeddings are quantised to int8 and bulk-indexed to Elasticsearch. Total pipeline execution (data

extraction $\rightarrow$ trained model $\rightarrow$ indexed embeddings) spans approximately 4 days wall-clock time.

5 Training Methodology

Symphony employs a three-phase curriculum that progressively builds from phonetic feature learning through knowledge distillation to discriminative fine-tuning.

5.1 Phase 1: Teacher Training on Phonetic Features

The Teacher network learns to produce embeddings where phonetically similar toponyms cluster together, using triplet margin loss with $m = 0.3$. To prevent learning trivial script-based boundaries, I implement script-aware negative sampling: 80% of negatives are drawn from the same writing system as the anchor, forcing fine-grained phonetic discrimination within scripts. Negative selection also enforces attestation disjointness (Section 4.3), preventing false negatives from homonyms.

From 8.4 million unique toponym IDs referenced in 65.1 million positive pairs, filtering yields **20.4 million training** and **2.3 million validation triplets**. The hybrid G2P strategy achieves 53.4% IPA coverage across 57.6M training-namespace toponyms.

5.2 Phase 2: Student-Teacher Alignment

The Student learns to approximate Teacher embeddings from character sequences, minimising MSE and cosine distance to frozen Teacher outputs. Language dropout (50%) forces learning of script-intrinsic patterns, while noise augmentation (30%) trains robustness to input corruption.

Handling Orthographic Twins. When the same string has different pronunciations across languages (e.g., “Paris”: French /paʁi/ vs. English /ˈpærɪs/), the Teacher provides ground truth via IPA-derived features, pushing embeddings apart or keeping them close accordingly. The Student receives identical character sequences but distinct language embeddings, learning when language context modifies phonetic interpretation.

Student-Teacher cosine similarity reached 0.943, indicating strong phonetic knowledge transfer from IPA-based to character-based representations.

5.3 Phase 3: Discriminative Fine-Tuning

While distillation aligns the Student with the Teacher’s general phonetic space, minimal pairs—names with high similarity but distinct phonetic realizations (e.g., “Austria” vs. “Australia”)—require sharper boundaries.

Hard Negative Mining. Triplets (a, p, n) are constructed where anchor a and positive p refer to the same place (via co-location in gazetteers), while hard negative n has high orthographic similarity to a (same 2-character prefix, same script) but shares no attestations with a (Section 4.3). This attestation disjointness constraint prevents false negatives from homonyms while accepting the limitation that cross-authority duplicates (e.g., the same city in both GeoNames and Wikidata under unlinked records) may occasionally be incorrectly paired.

Elasticsearch KNN queries with attestation filtering yield **8 million training** and **1 million validation triplets** from bin-balanced pairs (25k per script:language, down from 50k to prevent high-resource dominance).

Trade-off. Hard negatives are sampled from the same script, teaching finer discrimination within scripts at a modest cost to cross-script performance. Since Symphonym’s primary goal is cross-script matching (linking “北京” to “Beijing”), and same-script variants can be handled by traditional edit-distance methods, this trade-off is acceptable.

6 Evaluation

6.1 Embedding Quality

Table 2 summarises embedding quality across diagnostic categories after Phase 3 training.

Test Category	Pass Rate	Description
Cross-script equivalents	18/22 (81.8%)	Latin ↔ Cyrillic, Greek, Arabic, CJK, Hebrew, Hangul
Diacritic variants	4/4 (100%)	Zurich/Zürich, Krakow/Kraków, São Paulo/Sao Paulo
Unrelated pairs	3/3 (100%)	Correctly separated (low similarity)
Total	25/29 (86.2%)	

Table 2: Production embedding quality diagnostics. Cross-script matching is the primary design goal. Tested on 66.9M toponyms with 100% embedding coverage across 20 scripts.

Cross-script Equivalents. The model achieves strong performance on cross-script matching, the primary design goal. Production testing shows representative similarities: Latin-Cyrillic pairs achieve 0.94–0.99 (London/Лондон: 0.991, Berlin/Берлин: 0.988, Moscow/Москва: 0.945), Latin-Greek 0.96–0.98 (Athen-s/Aθήνα: 0.980), Latin-CJK 0.81–0.96 (Beijing/北京: 0.955, Shanghai/上海: 0.945), Latin-Arabic 0.91–0.97 (Baghdad/بغداد: 0.969), and Latin-Hebrew 0.89–0.91 (Jerusalem/ירושלים: 0.892).

Same-script Cross-language: Correct Phonetic Discrimination. Cross-language same-script pairs with genuinely different pronunciations receive appropriately low similarity scores, demonstrating the model’s correct phonetic discrimination rather than a deficiency. London/Londres (0.474) and London/Londra (0.574) reflect real phonetic differences: English /ˈlʌndən/ vs French /lɔ̃dʁ/ vs Italian /ˈlɒndra/. These are not transliterations but exonyms with distinct pronunciations that should be distinguished.

In the WHG reconciliation pipeline, such phonetically distinct variants are linked through place-level attestations (co-occurrence in authority records like GeoNames or Wikidata) rather than phonetic similarity. Symphonym correctly identifies phonetic equivalence for cross-script pairs (London/Лондон: 0.991) where pronunciation is preserved despite orthographic transformation, while appropriately separating phonetically divergent forms within the same script. This behaviour enables the system to surface high-confidence phonetic matches for cross-script scenarios while avoiding false positives from orthographically similar but phonetically distinct exonyms.

Historical Variants. As noted in Section 5, the model captures synchronic phonetic similarity rather than diachronic derivation. Pairs like Istanbul/Constantinople (0.719) and Paris/Lutetia (0.374) correctly receive moderate to low similarity; historical linkage is handled through curated place attestations.

Script-Pair Analysis. Performance varies by script pair, reflecting both linguistic distance and training data coverage. Latin↔Cyrillic pairs achieve high similarity (mean 0.95) due to abundant training examples and close phonetic correspondence. Latin↔CJK pairs perform strongly (mean 0.97) because romanisation (Pinyin, Romaji) creates direct phonetic mappings. More challenging are pairs involving scripts with complex orthography-phonology relationships: Latin↔Arabic achieves 0.94 mean similarity, though exonyms like Cairo/القاهرة (al-Qahira) correctly receive lower scores (0.62) reflecting genuine phonetic divergence.

6.2 MEHDIE Benchmark

I evaluate Symphonym on the MEHDIE Hebrew-Arabic historical toponym benchmark (Sagi et al., 2025) using ranking metrics (Recall@K, MRR) that reflect the system’s design as a candidate generator for manual review workflows rather than a threshold-based classifier. Table 3 presents comparative performance across all five testsets.

Ranking Performance. Symphonym achieves the strongest Recall@1 performance (86.7%) among all methods, outperforming both Levenshtein (81.5%) and Jaro-Winkler (78.5%). This demonstrates that Symphonym’s phonetic embeddings rank the correct match highest more consistently than character-based methods.

Method	Testset	R@1	R@5	R@10	R@20	MRR
Levenshtein	TS7 (Yaqut-Kima Sham)	75.8	93.9	100.0	100.0	83.7
	TS8 (Kima-Thurayya Sham)	90.5	100.0	100.0	100.0	94.4
	TS9 (Tudela-Thurayya)	77.8	100.0	100.0	100.0	87.5
	TS10 (Yaqut-Kima Maghreb)	66.7	97.0	97.0	97.0	79.6
	TS11 (Damast-Tudela)	96.9	96.9	100.0	100.0	97.3
	Mean	81.5	97.5	99.4	99.4	88.5
Jaro-Winkler	TS7 (Yaqut-Kima Sham)	75.8	100.0	100.0	100.0	86.4
	TS8 (Kima-Thurayya Sham)	90.5	90.5	95.2	100.0	91.4
	TS9 (Tudela-Thurayya)	77.8	100.0	100.0	100.0	86.6
	TS10 (Yaqut-Kima Maghreb)	54.5	93.9	97.0	100.0	71.9
	TS11 (Damast-Tudela)	93.8	96.9	96.9	96.9	95.4
	Mean	78.5	96.2	97.8	99.4	86.3
Symphonism	TS7 (Yaqut-Kima Sham)	81.8	93.9	97.0	97.0	87.2
	TS8 (Kima-Thurayya Sham)	90.5	100.0	100.0	100.0	93.0
	TS9 (Tudela-Thurayya)	94.4	100.0	100.0	100.0	97.2
	TS10 (Yaqut-Kima Maghreb)	69.7	78.8	90.9	93.9	75.8
	TS11 (Damast-Tudela)	96.9	100.0	100.0	100.0	98.4
	Mean	86.7	94.5	97.6	98.2	90.3

Table 3: MEHDIE benchmark ranking metrics (all values in %). Symphonism achieves the highest Recall@1 (86.7%) and MRR (90.3%), demonstrating superior ranking quality for candidate generation workflows. R@10 of 97.6% means correct matches appear in the top 10 candidates over 97% of the time. Levenshtein achieves highest R@10 (99.4%) but lower R@1, placing correct matches further down the ranked list.

The Mean Reciprocal Rank of 90.3% confirms that correct matches typically appear in position 1, making manual review highly efficient.

At Recall@10, Symphonism achieves 97.6%—only slightly below Levenshtein’s 99.4% but substantially ahead of Jaro-Winkler’s 97.8%. This performance profile reflects Symphonism’s design goals: prioritise placing correct matches at the top of the ranked list (high R@1, high MRR) rather than ensuring they appear somewhere in a longer list (R@20). For a gazetteer reconciliation workflow where users review top-ranked candidates, R@1 and MRR are the critical metrics.

Testset-Specific Analysis. Performance varies by testset complexity:

- **TS9 (Tudela-Thurayya) and TS11 (Damast-Tudela):** Symphonism achieves near-perfect performance (R@1: 94.4%–96.9%, R@10: 100%), demonstrating strong handling of these medieval-to-modern Arabic mappings.
- **TS8 (Kima-Thurayya Sham):** All three methods achieve 90.5% R@1, with Symphonism matching baseline performance.

- **TS7 (Yaqut-Kima Sham):** Symphony’s 81.8% R@1 exceeds baselines (75.8%), though R@10 is slightly lower (97.0% vs 100.0%).
- **TS10 (Yaqut-Kima Maghreb):** This proves most challenging for all methods. Symphony achieves 69.7% R@1 (vs Levenshtein 66.7%, Jaro-Winkler 54.5%) but lower R@10 (90.9% vs 97.0%). The Maghreb toponyms contain more phonetically divergent historical variants, reflecting genuine pronunciation evolution rather than simple transliteration differences.

Comparison with MEHDIE System. The MEHDIE paper (Sagi et al., 2025) reports results using threshold-based F-5 metrics, making direct comparison with ranking metrics difficult. Their specialised system achieves F-5=0.76 (mean across testsets) through explicit phonetic transcription via Phonetisaurus G2P models optimised for medieval Arabic-Hebrew matching. Symphony’s architecture differs fundamentally: rather than explicit IPA conversion, it learns implicit cross-script phonetic representations from modern multilingual gazetteer data, prioritizing generalization across 20 scripts over optimization for specific historical language pairs.

The approaches are complementary:

- **MEHDIE** uses language-pair-specific G2P models with explicit phonetic conversion at query time, achieving high precision on the medieval Semitic variants for which it was specifically tuned.
- **Symphony** maps characters directly to a learned phonetic embedding space without runtime phonetic conversion, operating efficiently at scale across diverse script combinations without requiring language-specific resources for each new pair.

Both systems substantially outperform simple romanisation approaches, confirming that phonetic similarity (whether explicit or learned) is essential for cross-script toponym matching.

6.3 Production Deployment Test Results

To validate production readiness, I deployed the model’s outputs to a staging Elasticsearch instance containing all 66,924,548 toponyms from the WHG corpus and executed a comprehensive test suite measuring embedding coverage, cross-script similarity, and retrieval quality.

Embedding Coverage. The production index achieves 100% embedding coverage: all 66.9M toponyms have valid Symphony embeddings.

Cross-script Similarity Validation. To comprehensively evaluate cross-script matching capability, I sampled 12,947 genuine cross-script toponym pairs from the v6 training data (systematically drawing up to 10 samples from each of 1,366 cross-script language-script bins). These pairs represent the actual distribution of cross-script equivalents the model encountered during training. Testing yields 82.6% pass rate (10,697/12,947 pairs exceed the somewhat-arbitrary 0.75 similarity threshold), with mean similarity 0.879 and median 0.945.

Similarity Distribution: 47.6% of successful matches achieve ≥ 0.95 (excellent); 18.8% fall in 0.90–0.95 (good); 14.8% in 0.80–0.90 (fair); 7.5% in 0.70–0.80 (acceptable); and 11.4% below 0.70 (poor but above noise floor). The high concentration at ≥ 0.95 confirms that most cross-script equivalents are strongly clustered in embedding space.

Performance by Script Combination: Best-performing combinations include Arabic-Cyrillic (94–100% pass rate across 5 language pairs), Cyrillic-Hebrew (100%, $n=50$), Arabic-Latin major languages (100% for Arabic-English, Arabic-German, Arabic-Spanish), Cyrillic-Latin (94.3%, $n=1,306$), and Gujarati-Latin (96.9%, $n=229$). Problematic combinations include CJK-Latin (75.0%, $n=1,071$), CJK-Hangul (69.8%, $n=63$), and Japanese Kana scripts (Hiragana cross-script: 7.8%; Katakana cross-script: 20.3%). The Kana results reflect the routing bug described in §9, though partial phonetic learning through cross-script generalisation yields better-than-random performance.

Representative high-quality matches demonstrate cross-script generalisation:

- مطار كاليكوت الدولي (ar) / কালিকট আন্তর্জাতিক বিমানবন্দর (bn): 0.987 (Arabic-Bengali)
- كورستانكو (ar) / Користанко (ru): 0.984 (Arabic-Cyrillic)
- London / Лондон (Latin-Cyrillic): 0.991
- Athens / Αθήνα (Latin-Greek): 0.980
- Beijing / 北京 (Latin-CJK): 0.955
- Jerusalem / ירושלים (Latin-Hebrew): 0.892

The systematically sampled test confirms that strong cross-script performance generalises beyond manually selected capitals to the full diversity of place names in the training corpus.

Diacritic Robustness. All four diacritic variant tests pass with similarities ≥ 0.95 (Zurich/Zürich: 0.979, Kraków/Krakow: 0.952, São Paulo/Sao Paulo: 0.984, Bogotá/Bogota: 0.975), confirming that the model correctly treats diacritics as phonetic modifications rather than character substitutions.

Embedding Quality Metrics. Analysis of 10,000 randomly sampled embeddings confirms proper L2 normalisation (mean norm: 1.000 ± 0.002) and good discriminative power (mean pairwise similarity: 0.059 ± 0.305 , median: 0.063). The low mean similarity and wide range $[-0.95, 0.97]$ demonstrate that embeddings successfully separate unrelated toponyms while clustering phonetically similar ones.

KNN Retrieval Evaluation. The KNN test evaluates whether Symphonium can retrieve expected cross-script and cross-language variants when querying with major city names. Unlike pairwise similarity tests (which directly compare known equivalents), KNN retrieval tests whether the model’s embedding space naturally clusters variants together when performing open-ended searches across the full 67M toponym corpus.

Testing on three representative queries (London, Paris, Moscow) yields mixed results that illuminate both capabilities and limitations:

- **London (en):** Successfully retrieves Лондон (Cyrillic, 0.9967 similarity), correctly excluding phonetically distinct Romance variants (Londres, Roma). Also surfaces Arabic لندون (0.9908) and Georgian ლონდონი (0.9889)—demonstrating cross-script generalization.
- **Paris (fr):** Does not retrieve the Romance variants with different pronunciations (Parigi Italian, París Spanish), but does surface cross-script equivalents: Парис (Cyrillic, 0.9945), Thai ปารีส์ (0.9962), and Greek Πάρις (0.9926).
- **Moscow (en):** Retrieves Moscou (French/Portuguese, 0.9945)—a phonetically similar variant—and Москва (Cyrillic, 0.9941). Also surfaces the data quality issue “Kachua-mokampukur F P School” (0.9874), an OpenStreetMap (OSM) institutional name with implausible en/nl language tags.

The results suggest **two distinct phenomena**:

(1) Script family clustering is robust. The model successfully identifies cross-script equivalents in Cyrillic, Arabic, Thai, and Georgian—scripts represented in the training data. The London query’s retrieval of Arabic and Georgian variants demonstrates genuine phonetic generalisation, correctly distinguishing these transliterations from phonetically distinct Romance forms.

(2) Romance variants reflect genuine phonetic differences. The absence of Spanish *Londres*, Italian *Parigi*, or Portuguese *Moscou* from top KNN results is not a model deficiency but correct phonetic behaviour: these forms represent genuine pronunciation differences, not mere transliterations. English “London” /ˈlʌn.dən/ differs significantly from French “Londres” /lɔ̃dʁ/ (nasal vowel, uvular /ʁ/); Italian “Roma” /ˈro.ma/ differs from English “Rome” /rʊm/ (diphthong vs. pure vowel). The pairwise similarities (Rome/Roma: 0.788, Munich/München: 0.878) reflect these phonetic distances, not embedding failures. In the World Historical Gazetteer (WHG) reconciliation pipeline, such variants are linked at

the *place* level—users querying for London-related places would retrieve records linked to the London place entity, including its French, Spanish, and Italian attestations—rather than at the toponym similarity level.

The OSM institutional name retrieval (“Kachua-mokampukur F P School” matching “Moscow” at 0.9874) reveals the limitation of PanPhon192’s fixed 8-bin positional pooling. With short strings (3-8 characters), each bin captures a meaningful phonetic segment, producing sharp, distinctive representations. With long strings like the 35-character compound name, phonetic features are averaged across many segments per bin, creating a “smoothed” representation where local phonetic subsequences (/mok/ in “mokampukur”) can accidentally align with query bins containing /mosk/ in “Moscow”. This is a direct consequence of the fixed bin count: longer names dilute positional specificity, reducing discriminative power. This is discussed further in Section 7.3.

Implications for production deployment: KNN retrieval on real gazetteer data must account for high-multiplicity clusters (“London” appears in 69 language variants with identical embeddings, dominating top-k results). Post-processing strategies include: query-time filtering (exclude exact matches to surface cross-script alternatives), script diversity re-ranking, or candidate expansion (retrieve top-100, then filter/rank by additional constraints). The test validates Symphonym as a *candidate retrieval* mechanism rather than a deterministic ranker: embeddings provide phonetically-informed similarity scores, while downstream components apply geographic and entity-type constraints to select among candidates.

Production Implications. The 100% embedding coverage and strong cross-script performance confirm that the hybrid G2P pipeline (Epitrans + 102 extensions + Phonikud + CharsiuG2P) successfully addresses all previously identified coverage gaps. The production corpus of 66.9M embedded toponyms enables phonetically-informed search across the full WHG corpus, supporting cross-script reconciliation workflows without requiring language-specific resources or runtime phonetic conversion.

7 Discussion

The evaluation results support Symphonym’s central claim: that a character-level neural encoder can learn phonetically meaningful representations across writing systems without requiring explicit phonetic transcription at inference time.

7.1 Key Findings

Version 6 Improvements. The transition from v5 to v6 addresses the critical coverage gaps identified during v5 evaluation. The hybrid G2P strategy—combining 100 new Epitrans extension files (developed using multi-LLM cross-checking, without human curation), Phonikud for Hebrew, and CharsiuG2P for

CJK topolects—expands IPA coverage from 50.8% to 53.4% (30.8M of 57.6M training namespace toponyms). The elimination of AnyAscii romanisation for CJK scripts means the Teacher now learns from genuine phonetic representations rather than lossy approximations: Chinese /bei.dʒɪŋ/ rather than romanised “bei-jing”, Hebrew /jəruʃaˈlajim/ from unpointed ירושלים rather than absent embeddings. Previously zero-coverage scripts now show substantial IPA generation: Greek (77.4%), Armenian (93.7%), Gujarati (94.9%), Kannada (48.6%). Phase 1 training on the expanded v6 dataset shows val_loss consistently $\sim 1.23\times$ higher than v5 at equivalent epochs (e.g., 0.0072 vs 0.0054 at epoch 10), reflecting the harder but more linguistically faithful discrimination task. Despite fewer training samples (20.4M vs 24.8M for v5, due to tighter bin-balancing caps), v6’s improved IPA coverage and elimination of romanisation artifacts successfully address script-specific performance gaps observed in v5 MEHDIE testing.

Cross-script Generalisation. Comprehensive testing on 12,947 systematically sampled cross-script pairs from the training data achieves 82.6% pass rate at the 0.75 similarity threshold (mean 0.879, median 0.945), with 47.6% of successful matches exceeding 0.95 (excellent quality). Best-performing script combinations include Arabic-Cyrillic (94–100%), Cyrillic-Latin (94.3%), and Gujarati-Latin (96.9%), demonstrating robust generalisation across historically disconnected writing systems. The test sampled up to 10 pairs from each of 1,366 cross-script language-script bins, ensuring coverage of the full training distribution rather than cherry-picked examples. On the MEHDIE Hebrew-Arabic benchmark of medieval toponyms, the system achieves Mean Reciprocal Rank of 0.903 and Recall@10 of 97.6%, meaning the correct cross-script match appears in the top 10 candidates over 97% of the time. KNN retrieval tests across the full 67M-name corpus reveal a practical challenge: when querying for major cities, the system successfully ranks cross-script variants within top results, but must compete with many Latin-script variants and compound forms (“Municipality of Athens”, “Prefecture de Atenas”) that also score highly. This reflects not an embedding quality problem—pairwise tests show strong similarity scores—but rather the density of related names in a production gazetteer. The MEHDIE evaluation is particularly notable because the benchmark sources (medieval Hebrew and Arabic geographical texts) are independent of the modern gazetteers used for training. The strong cross-temporal performance suggests the model has learned general cross-script phonetic mappings rather than memorising specific toponyms. This generalisation emerges from the Teacher-Student architecture: the Teacher learns language-specific phonetic mappings from IPA features, then transfers this knowledge to the Student through distillation.

Retrieval vs. Classification. Symphonism excels at *ranking* candidates by phonetic similarity but is not optimised for *classification* at fixed thresholds. The MEHDIE benchmark comparison illustrates this: when evaluated with ranking

metrics (Recall@10: 97.6%, MRR: 0.903), Symphonium substantially outperforms baselines; threshold-based metrics (F-5) favour methods tuned for specific cutoff points. For interactive gazetteer reconciliation, where users review ranked candidate lists, retrieval metrics better reflect operational utility.

Same-script Performance and Phonetic Fidelity. Same-script cross-language pairs achieve 86% pass rate, comparable to cross-script pairs (93%). This reflects the model’s correct phonetic discrimination: exonyms like London/Londres (0.474) or Roma/Rome (0.788) have genuinely different pronunciations (English /ˈlʌndən/ vs French /lɔ̃dʁ/; Italian /roːma/ vs English /roʊm/) and should receive lower similarity scores. The model distinguishes between two scenarios: (1) transliterations that preserve pronunciation across scripts (London/Лондон: 0.991), where high similarity is appropriate; and (2) phonetically distinct exonyms within the same script (London/Londres: 0.474), where low similarity correctly reflects the phonetic divergence.

This behaviour is operationally desirable: in the WHG reconciliation pipeline, phonetically divergent same-script variants are linked through place-level attestations (co-occurrence in authority records) rather than phonetic similarity. The model’s strong cross-script performance combined with conservative same-script scoring minimises false positive matches while ensuring that true phonetic equivalents (cross-script transliterations) are reliably detected. For same-script refinement within a single language, traditional string methods (Levenshtein, Jaro-Winkler) remain effective and complement Symphonium’s cross-script capabilities.

Hard Negative Training. Phase 3 training sharpens discrimination between orthographically similar names at a modest cost to same-script cross-language pairs. Validation during training showed that cross-script pairs improved consistently: Thessaloniki/Θεσσαλονίκη rose from 0.848 to 0.963 similarity by epoch 10, while same-script pairs like London/Londra declined slightly (0.617 to 0.598). This is expected: hard negatives are sampled from the *same* script, teaching the model to be more discriminating within scripts. Since Symphonium’s primary goal is cross-script matching—linking “北京” to “Beijing” rather than distinguishing “London” from “Londres”—this trade-off is acceptable. In the WHG system, same-script variants remain linked at the place level through co-location in gazetteer records. Moreover, same-script matching can be effectively handled by traditional string similarity methods (Levenshtein distance, Jaro-Winkler) that complement Symphonium’s cross-script capabilities; a hybrid pipeline might apply Symphonium for cross-script candidate retrieval and fall back to edit-distance methods for same-script refinement.

7.2 Operational Implications

Scalability and Deployment. Symphonism embeddings integrate directly with Elasticsearch’s HNSW approximate nearest-neighbour indexing, enabling sub-second retrieval over the full WHG corpus of 67M toponyms. Query latency averages 15–50ms including network overhead, making interactive reconciliation workflows practical. Unlike rule-based phonetic algorithms (Soundex, Metaphone), which require exact code matches, embedding similarity enables fuzzy matching with graceful degradation. The Student model’s inference cost is minimal: encoding a toponym requires a single forward pass through an 8.3M-parameter network, achievable in <1ms on CPU. This operational efficiency—the fact that Symphonism *actually runs* at scale with acceptable latency—is a key contribution beyond the academic novelty of the embedding approach.

No Runtime Phonetics. The Student encoder requires only raw character input—no language identification, no G2P conversion, no phonetic databases. This simplifies deployment and eliminates failure modes associated with unknown languages or unsupported scripts.

Complementary Methods. Symphonism is designed as one component of a larger reconciliation pipeline. Cross-script candidate retrieval benefits from phonetic embeddings; same-script refinement may use edit distance; final disambiguation incorporates geographic proximity and temporal constraints. The embedding similarity score provides a phonetic prior that downstream components can weigh against other evidence.

Historical Orthography. The model’s ability to cluster pre-standardisation spelling variants—the original motivation for phonetic embeddings in Section 7—is empirically validated. Medieval names exhibit exactly the kind of phonetically consistent but orthographically variable variation that Symphonism was designed to handle.

7.3 Limitations

Several limitations warrant acknowledgment:

Training Data Bias. Despite the stratified sampling strategy described in Section 4.1, training sources retain biases. GeoNames over-represents populated places with official names, potentially under-representing historical variants. Wikidata’s coverage skews toward places of encyclopaedic interest. TGN emphasises art-historically significant locations. Performance on under-represented scripts (e.g., Ethiopic, Myanmar, Tibetan) and mundane places lacking multi-lingual attestations may be weaker.

Tonal Languages. The current architecture does not explicitly model tone, which is phonemically contrastive in Chinese, Vietnamese, Thai, and other languages. The PanPhon features (Section 3.5) encode segmental articulatory properties but not suprasegmental features. In principle, tonal minimal pairs in place names would not be distinguished. However, tonal minimal pairs are rare in practice: Chinese place names like Beijing (北京, běijīng) and Shanghai (上海, shànghǎi) have distinct segmental content that Symphonism captures effectively (0.97–0.99 similarity with romanised forms). The limitation would matter most for hypothetical pairs differing only in tone, which are uncommon in geographic naming.

IPA Coverage. The Teacher’s phonetic grounding depends on Epitran’s G2P coverage. Of 57.6M training toponyms, 30.8M (53.4%) yielded valid IPA transcriptions and PanPhon embeddings; the remainder contribute only through Phase 2 and 3 character-level training. Coverage varies significantly by script (see Table 1). The v6 hybrid G2P strategy achieves strong coverage across multiple script families: South Asian scripts (Tamil 90.9%, Telugu 92.6%, Bengali 72.9%) via Epitran’s Indic language support; Greek (77.4%), Armenian (93.7%), and Kannada (48.6%) via extended Epitran; Hebrew (83.8%) via Phonikud; and CJK (50.1%) and Korean (58.0%) via CharsiuG2P. The strong performance on Georgian (Tbilisi/თბილისი = 0.972) despite Georgian representing only 0.2% of training data suggests the Student learns effective representations through character-level patterns and cross-script transfer.

Confusable Pairs. The model correctly assigns high similarity to phonetically similar strings regardless of semantic relationship. Pairs like Austria/Australia (0.883) and China/Ghana (0.932) receive high scores because they *are* phonetically similar. Disambiguation requires geographic or contextual evidence beyond phonetic similarity.

Length Sensitivity. The 8-bin positional pooling in PanPhon192 (Section 4.2) produces fixed-length representations regardless of input length, creating an asymmetry: short names occupy bins with high phonetic specificity, while long names average features across many segments, reducing discriminative power. This affects both the Teacher’s embedding quality for long strings and the Student’s ability to learn length-appropriate similarity judgments. The concrete example from KNN retrieval (Section 6)—where “Kachua-mokampukur F P School” (35 characters) matches “Moscow” (6 characters) at 0.987 similarity due to local phonetic subsequence alignment—provides empirical evidence of this limitation. Short toponyms (3–8 characters) produce sharp, distinctive bin representations with each bin capturing meaningful phonetic segments, while long compound names or institutional strings produce smoothed representations where distinct subsequences can accidentally align with unrelated short queries.

8 Application to Medieval Personal Name Matching

While designed for toponyms, Symphonym’s phonetic embeddings transfer effectively to personal names in historical contexts lacking orthographic standardisation. A case study from the Medieval London Customs Accounts project (University of London) demonstrates this capability: merchants’ names appear with variant spellings (“Cristofre”/“Christopher”, “Arnaldson”/“Arnaldisson”) reflecting phonetic transcription rather than orthographic rules. Within a supervised active learning framework, Symphonym embeddings computed from concatenated forename+surname strings (e.g., “John Arnaldson”) serve as features alongside traditional string metrics, enabling prosopographic analysis and network reconstruction. The system successfully clusters phonetically similar name variants without retraining, indicating the learned representations capture general orthography-phonology relationships beyond the toponym domain. This demonstrates broader applicability: any scholarly domain involving historical documents with inconsistent orthography—prosopography, onomastics, archival indexing—may benefit from phonetically grounded embeddings to complement traditional record linkage methods.

9 Future Work

Japanese Kana Scripts (Hiragana/Katakana). Symphonym v6 exhibits 0% IPA coverage for Hiragana (151,980 toponyms) and Katakana (340,555 toponyms), despite both scripts being natively supported by EpiTran via `jpn-Hira` and `jpn-Kana` language codes. The root cause is a routing error: the pipeline dispatches by language first (`lang=ja`), directing all Japanese toponyms to `CharsiuG2P`, which only processes CJK characters (Kanji). Hiragana and Katakana are consequently ignored.

Testing on 333 real Japanese Kana pairs from v6 training data reveals partial success despite this gap: 23% (77/333) achieve similarity scores above 0.70, with many more in the 0.64–0.81 range. Cross-script pairs like さつぽろ/سابورو (similarity 0.789) demonstrate that the Student network learned Japanese phonetic patterns indirectly through Phase 3 hard-negative mining. This graceful degradation—rather than catastrophic failure—suggests the model captures phonetic relationships even without explicit Teacher supervision.

Version 7 will implement script-first routing to restore phonetic targets for these 492,535 toponyms (0.7% of corpus), expected to improve the 23% pass rate on Kana pairs to levels comparable with properly supervised scripts (typically >90%).

Length-Aware Embeddings. Incorporating string length as an explicit conditioning signal in v7—either as a direct input feature to the encoder or as a learned length embedding concatenated with character embeddings—would en-

able the model to learn that similarity scores should be interpreted relative to input lengths, penalising matches between strings of very different lengths. This addresses the length sensitivity limitation identified in Section 7.3, where PanPhon192’s fixed 8-bin positional pooling creates asymmetric representation quality for short versus long names. An alternative or complementary approach would be to modify the PanPhon192 pooling strategy itself: adaptive bin counts scaled to input length, or hierarchical pooling that preserves local phonetic structure for longer strings. Both approaches would reduce spurious high-similarity matches between short toponyms and long compound names or institutional strings, improving precision in real-world retrieval scenarios without sacrificing the fixed-dimensionality property essential for efficient vector search.

10 Conclusion

Symphonym maps names from 20 writing systems into a unified 128-dimensional phonetic space, enabling cross-script matching without runtime phonetic conversion. The three-phase curriculum—phonetic feature learning, teacher-student distillation, and hard negative fine-tuning—transfers knowledge from IPA-derived articulatory features to a character-level model requiring no phonetic resources at inference. Trained on 67M toponyms from GeoNames, Wikidata, and Getty TGN using hybrid G2P (Epitrans + 102 extensions, Phonikud, CharsiuG2P), the system achieves 86% accuracy on cross-script diagnostics and Recall@10 of 97.6% (MRR 0.903) on the MEHDIE Hebrew-Arabic historical benchmark.

The approach addresses practical challenges in digital humanities and geographic information retrieval: linking name references across languages, scripts, and historical periods without language-specific rules. By learning from multi-script toponyms in linked authority datasets, Symphonium captures phonetic correspondences difficult to enumerate manually. Fixed-length embeddings enable efficient KNN search in production environments.

Symphonym will be deployed in the World Historical Gazetteer’s reconciliation pipeline (2026), facilitating phonetic search for digital humanities and GLAM professionals. Beyond toponyms, a medieval personal names case study demonstrates effective transfer to historical orthographic variation, with implications for prosopographic analysis and archival research workflows requiring rapid similarity comparison across variable-length name strings.

Data Availability

All trained models, vocabularies, Epitrans extension files, and evaluation results supporting this study are openly available on Zenodo at <https://doi.org/10.5281/zenodo.18682017>.

The training data are derived from publicly available sources: GeoNames (<https://www.geonames.org/>), Wikidata (<https://www.wikidata.org/>), and the Getty

Thesaurus of Geographic Names (<https://www.getty.edu/research/tools/vocabularies/tgn/>). The MEHDIE benchmark used for evaluation is available via the supplementary file link at <https://link.springer.com/article/10.1007/s10579-025-09812-9>.

Code Availability

Training code and inference utilities are available at:

<https://github.com/WorldHistoricalGazetteer/indexing>.

The repository includes technical documentation of the HDBSCAN clustering methodology used for positive pair generation. The system integrates with Elasticsearch for scalable deployment.

Statements and Declarations

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Competing Interests

The author has no relevant financial or non-financial interests to disclose.

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